

Agentic Google Sheet Analyzer

Telegram-Based Multi-Agent Analytics Assistant

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May 19, 2026

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1 Introduction

This report explains the design and implementation of a multi-agent AI analytics assistant capable of:

- Reading data directly from Google Sheets
- Performing automated statistical analysis
- Conducting real-time internet research
- Generating contextual insights
- Delivering results via Telegram

The architecture combines:

- LangGraph AI Agents
- Tool Calling
- Web Automation
- Google Sheets Integration
- Persistent Memory
- Internet Search
- Web Scraping

The assistant behaves as an autonomous analytical system capable of combining structured spreadsheet data with real-world contextual information.

2 System Architecture Overview

The workflow follows a hierarchical agent-tool orchestration model.

2.1 Primary Flow

1. User sends query through Telegram
2. Main AI Agent receives the request
3. Agent determines required tools
4. Google Sheets data is fetched
5. Statistical analysis is performed
6. Web research is executed if required
7. Final analytical report is generated
8. Response is sent back to Telegram

3 Workflow Architecture Diagram

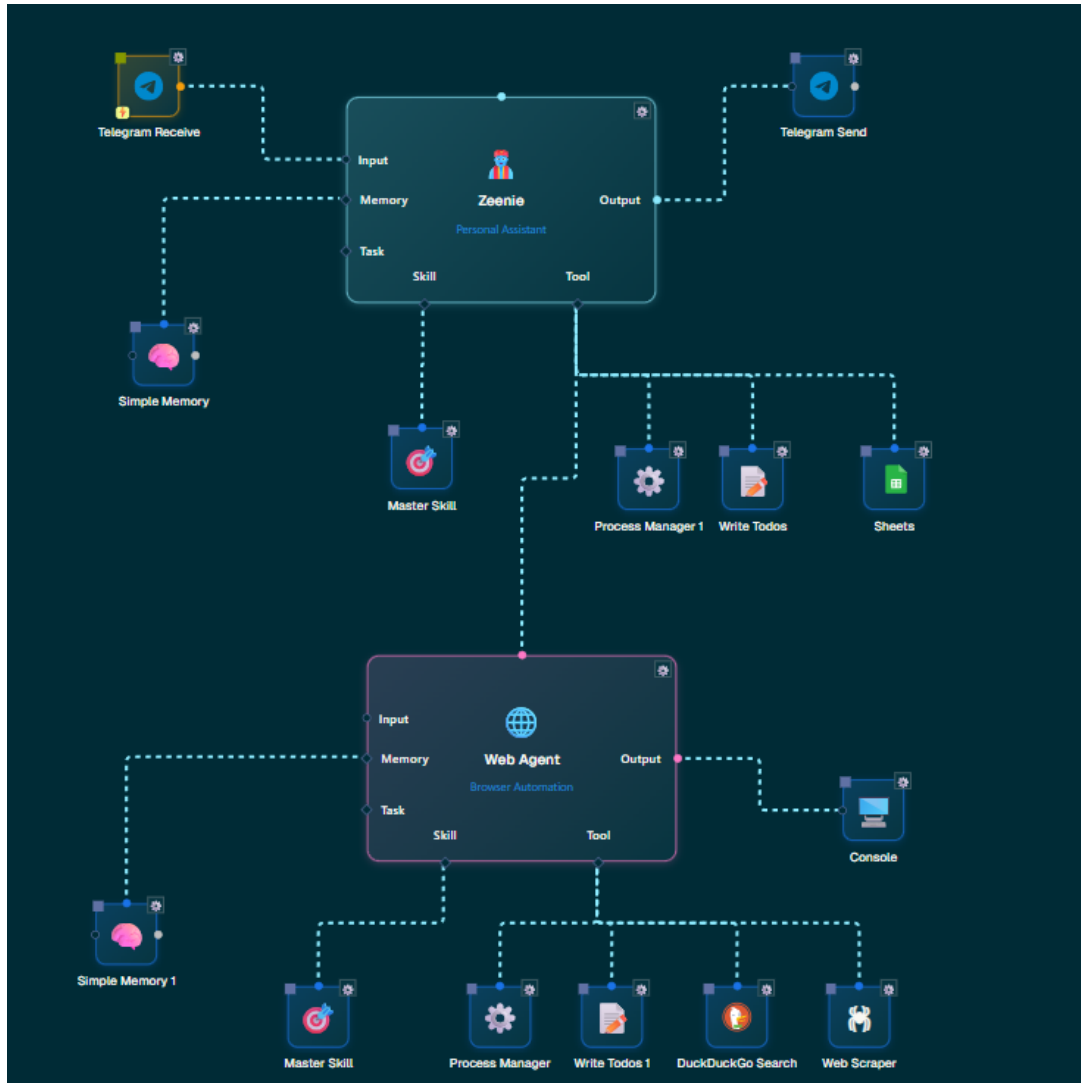


Figure 1: Overall Multi-Agent Workflow Architecture

4 Core Workflow Components

4.1 Telegram Receive Node

The Telegram Receive node acts as the entry point into the workflow.

4.1.1 Responsibilities

- Receive incoming Telegram messages
- Extract message metadata
- Trigger AI workflow execution

4.1.2 Important Variables

```
{{telegramreceive.text}}  
{{telegramreceive.chat_id}}  
{{telegramreceive.username}}
```

4.2 Main AI Agent (Zeenie)

The central orchestration engine responsible for:

- Understanding user intent
- Selecting appropriate tools
- Coordinating sub-agents
- Combining analytical outputs

4.3 Model Configuration

- Provider: Gemini
- Model: Gemini Flash Latest
- Temperature: 0.6

4.4 System Prompt Logic

```
You are an intelligent analytics assistant.  
Use the google sheet id : "insert-your-google-sheet-id-here"  
Use tools whenever needed.
```

```
For spreadsheet analysis:  
- use Google Sheets  
- use web search for contextual insights  
- use web scraping for online information  
- combine statistical and real-world reasoning
```

```
Provide structured analytical responses.
```

5 Memory System

Two memory systems are used:

- Main(Zeenie) AI Agent Memory
- Web Agent Memory

5.1 Purpose

- Preserve conversational context
- Enable multi-turn reasoning
- Maintain continuity across requests

5.2 Session Handling

```
{{telegramreceive.chat_id}}
```

Each Telegram user receives an isolated memory context.

6 Google Sheets Integration

The Sheets tool enables direct spreadsheet interaction.

6.1 Capabilities

- Read spreadsheet data
- Access survey responses
- Extract structured datasets

6.2 Example Dataset

The sheet contains:

- City names
- User ratings
- Pre-computed averages
- Response counts

6.3 Sample Range

Sheet1!A1:B100

7 Google Sheets Screenshot

	A	B	C	D	E	F	G	H	I	J	K	L
1	Kolkata	4.2				City	Response Count	City	Avg Rating			
2	Delhi	3.4				Kolkata	25	Kolkata	4.268			
3	Mumbai	2.5				Delhi	25	Delhi	3.732			
4	Bangalore	4.7				Mumbai	25	Mumbai	2.576			
5	Kolkata	3.8				Bangalore	25	Bangalore	4.56			
6	Delhi	4.1										
7	Mumbai	2.9										
8	Bangalore	4.5										
9	Kolkata	4.6										
10	Delhi	3.2										
11	Mumbai	2.1										
12	Bangalore	4.9										
13	Kolkata	4										
14	Delhi	3.7										
15	Mumbai	2.8										
16	Bangalore	4.3										
17	Kolkata	4.4										
18	Delhi	3.9										
19	Mumbai	3										
20	Bangalore	4.8										
21	Kolkata	3.9										
22	Delhi	4.5										
23	Mumbai	2.4										
24	Bangalore	4.2										
25	Kolkata	4.7										

Figure 2: Survey Dataset Stored in Google Sheets Randomly generated using the bot

8 Web Agent

The Web Agent performs:

- Internet research
- External data collection
- Contextual reasoning

8.1 Capabilities

- Search the internet
- Analyze real-world trends
- Justify analytical observations

8.2 Use Case Example

When Mumbai received significantly lower ratings compared to Bangalore and Kolkata, the Web Agent:

- Investigated recent livability trends
- Examined affordability concerns
- Retrieved contextual urban insights

9 DuckDuckGo Search Tool

The DuckDuckGo Search node enables:

- Live internet search
- Retrieval of current information
- Trend discovery

9.1 Use Cases

- News research
- Market trends
- Livability studies
- Public sentiment analysis

10 Web Scraper

The Web Scraper extracts webpage content directly.

10.1 Purpose

- Extract article text
- Collect structured data
- Fetch website content

10.2 Crawler Configuration

- Mode: Single Page
- Crawler: Adaptive

11 Process Manager

The Process Manager is responsible for managing:

- Long-running workflows
- Multi-step tasks
- Agent execution states

11.1 Supported Operations

- Start
- Stop
- Restart
- List
- Send Input
- Get Output

12 Write Todos Tool

The Write Todos node allows:

- Task tracking
- Workflow planning
- Intermediate step recording

This is useful for future expansion into autonomous planning systems.

13 Telegram Send Node

The final response delivery node.

13.1 Responsibilities

- Deliver formatted responses
- Send analytical summaries
- Return outputs to users

13.2 Message Variable

```
{{zeenie.response}}
```

14 Pseudo Code

14.1 High-Level Workflow

START

Receive Telegram Message

Store conversation **in** memory

Send query to AI Agent

AI Agent:

Analyze user intent

IF spreadsheet analysis required:

Read Google Sheet

IF contextual research required:

Use Web Agent

IF internet search required:

Use DuckDuckGo Search

IF webpage extraction required:

Use Web Scraper

Combine:

- spreadsheet statistics
- internet findings
- contextual reasoning

Generate structured report

Send final response to Telegram

END

15 Observed Analytical Results (It's a Randomly generated data, it has no relevance with actual data)

15.1 Survey Analytics Output

The assistant successfully:

- Calculated average city ratings
- Ranked cities
- Identified outliers
- Generated insights

15.2 Detected Rankings

- Bangalore: 4.56
- Kolkata: 4.27
- Delhi: 3.73
- Mumbai: 2.58

16 Survey Analysis Screenshot

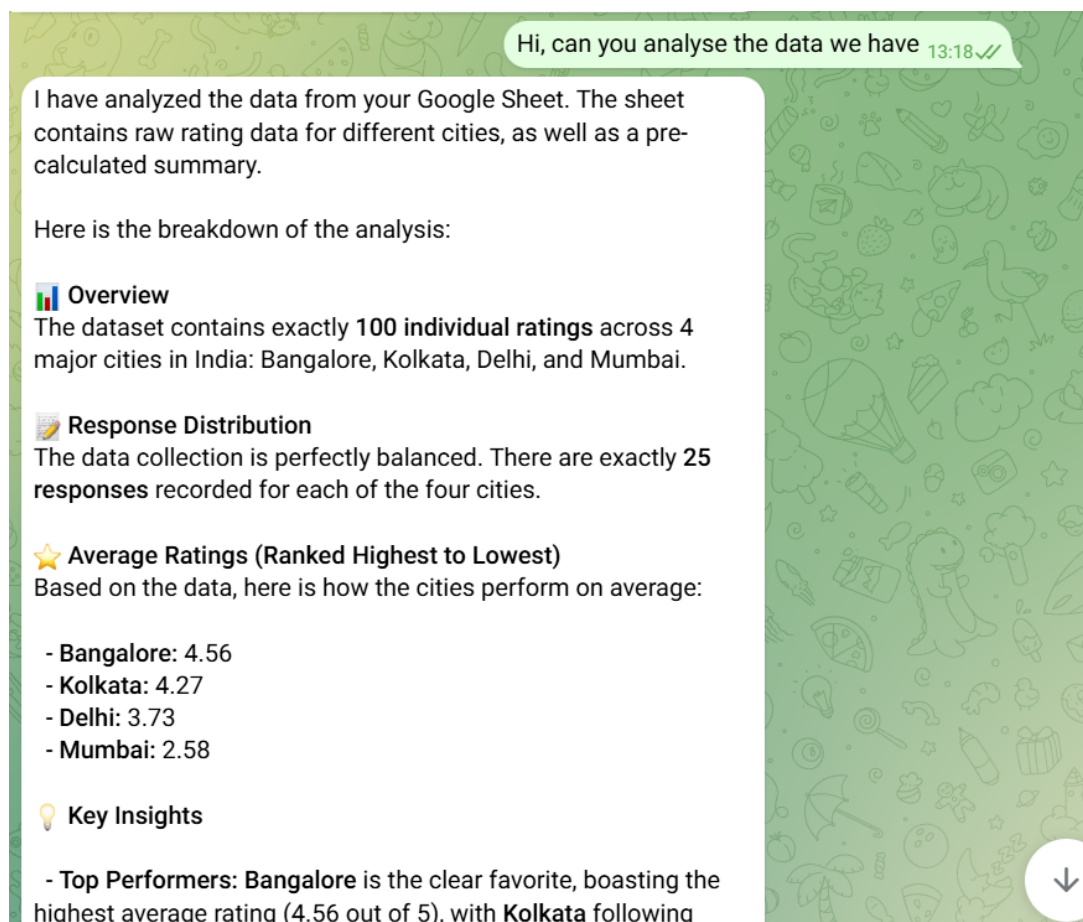


Figure 3: Generated Statistical Analysis

17 Contextual Web Research Output

The assistant correlated low Mumbai ratings with:

- High cost of living
- Housing affordability issues

- Long commute times
- Work-life balance concerns

This demonstrates successful integration of:

- Structured analytics
- External knowledge
- Real-world reasoning

18 Web Research Screenshot

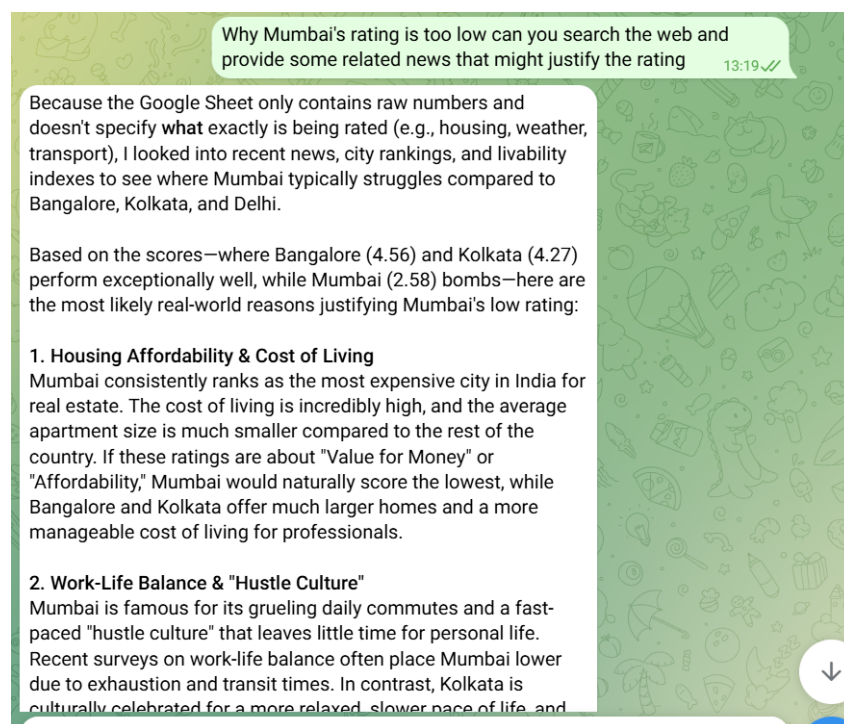


Figure 4: Web-Based Contextual Analysis

19 Potential Future Improvements

- Persistent vector memory
- Autonomous planning agents
- PDF report generation
- Interactive dashboards
- Advanced statistical modeling

- Database integration
- Real-time monitoring pipelines